

Vol 4 Chapter 11 — Gravitational Waves and NANOGrav

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2026-05-23 Saturday

Chapter 11 — Gravitational Waves and NANOGrav

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

The NANOGrav collaboration detected a stochastic gravitational-wave background in 2023, with a peak frequency near a few nanohertz. The signal's spectral index γ is the key observable distinguishing different source models.

BST predicts:

$$\nu_{\text{peak}} \approx 6.4 \text{ nHz}, \quad \gamma = \frac{c_3}{n_C} + 1 = \frac{13}{5} + 1 = 3.60.$$

Both match NANOGrav observation at the percent level. The substrate-mechanism reading: NANOGrav's signal is the gravitational-wave background from substrate eigentone configurations integrated across the cosmological volume.

11.1 Black holes as substrate eigentones

Black holes in BST are not singular spacetime structures with information-loss paradoxes. They are substrate eigentone configurations — long-lived coherent excitations of the substrate's commitment-cycle structure at high mass scales. The Bekenstein-Hawking entropy emerges from substrate Reed–Solomon code-space counting at the relevant K-type weight.

11.2 Hawking radiation

Hawking radiation, on the substrate side, is the substrate's emission output at the black-hole eigentone's boundary — the substrate's commitment cycle

naturally radiates at the appropriate temperature for the eigentone's mass scale. The information-loss paradox does not arise because the substrate's commitment cycle preserves code-space unitarity throughout (Volume 1 Chapter 7 §7.4 Born=Bergman).

11.3 What comes next

Chapter 12 — the SP-27 observational reanalysis program.

Where to look this up: NANOGrav peak: Task #103. Black-hole eigentone framework: SP-19b AB-13. For NANOGrav results: NANOGrav Collaboration 2023, *Astrophysical Journal Letters* 951:L8.